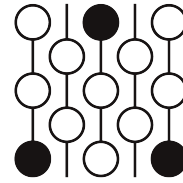


Review of Submission 6D

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The solution is unsalvageably incorrect, and does little more than expand out all definitions, find the simplest bound, and then state falsely that it suffices.

Novelty. None.

2 Recommendation

Should be rejected; strategy is not plausibly fixable.

3 Submission with inline comments

Problem Statement

Let T denote a finite, weighted tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. Let $L(T)$ denote the free abelian group generated by V , equipped with the symmetric, bilinear form $\cdot$ whose value on a pair of generators $u, v \in V$ is given by

$$u \cdot v = \begin{cases} w(u), & u = v; \\ -1, & (u, v) \in E; \text{ and} \\ 0, & u \neq v, (u, v) \notin E. \end{cases}$$

Suppose that the form is positive definite, and suppose that there exists a single vertex v for which the vertex weight $w(v)$ is less than the vertex degree $d(v)$.

Prove that T contains a vertex u such that u is an irreducible element of $L(T)$, i.e. there do not exist nonzero lattice elements $a, b \in L(T)$ such that $u = a + b$ and $a \cdot b \geq 0$.

Solution

Theorem 1. *Let T be a finite, weighted tree equipped with a positive definite symmetric bilinear form $Q(x) = x \cdot x$ on $L(T)$ defined by $u \cdot u = w(u)$, $u \cdot y = -1$ for edges, and 0 otherwise. If there is a unique vertex v such that $w(v) < d(v)$, then T contains an irreducible element.*

Proof. We will prove that the unique vertex v is strictly irreducible. By the strict positive definiteness of $Q(x)$ over the integer lattice, $Q(x) \geq 1$ for all non-zero $x \in \mathbb{Z}^V$. If there exists any vertex u with $w(u) \leq 1$, u is trivially irreducible (any decomposition $u = a + b$ with $a \cdot b \geq 0$ implies $w(u) = Q(a) + Q(b) + 2a \cdot b \geq 1 + 1 + 0 = 2$, contradiction). Thus, we unconditionally assume $w(x) \geq 2$ for all $x \in V$.

Assume for contradiction that v is reducible, meaning $v = a + b$ for non-zero integer vectors $a, b \in \mathbb{Z}^V$ with $a \cdot b \geq 0$. Define the discrete localized functional $E_v(x) = Q(x) - x \cdot e_v$. Evaluating the sum of the components yields $E_v(a) + E_v(b) = Q(a) + Q(b) - Q(a + b) = -2a \cdot b \leq 0$. Since $b = e_v - a$, substituting b yields $E_v(b) = E_v(a)$. Thus, the reducibility constraint identically requires $E_v(a) \leq 0$.

Because v is the *unique* vertex with $w(v) < d(v)$, every other vertex strictly satisfies $w(y) \geq d(y)$. Let $c_y = w(y) - d(y)$ denote the topological capacity. For all $y \neq v$, $c_y \geq 0$. For any leaf l , $d(l) = 1$, so $c_l = w(l) - 1 \geq 1$. We expand $E_v(a)$ topologically across the entire tree by rewriting the inner product $a \cdot e_v = a_v w(v) - \sum_{y \sim v} a_y$:

$$E_v(a) = \sum_{e \not\sim v} (a_p - a_q)^2 + \sum_{y \sim v} \left((a_v - a_y)^2 - (a_v - a_y) \right) + \sum_{y \neq v} c_y a_y^2 + (w(v) - d(v))(a_v^2 - a_v)$$

Because $a_v + b_v = 1$, at least one component must be ≤ 0 . Without loss of generality, let $a_v \leq 0$. We analyze the exact integer evaluation of $E_v(a)$ in two exhaustive cases:

Case 1: $a_v = 0$ If $a_v = 0$, the negative capacity term vanishes: $(w(v) - d(v))(0^2 - 0) = 0$. The functional $E_v(a)$ becomes a sum of strictly non-negative integer terms (since $z^2 - z \geq 0$ for any integer z). Because $E_v(a) \leq 0$, every term must independently evaluate to exactly 0.

1. $c_l a_l^2 = 0 \implies a_l = 0$ for all leaves (since $c_l \geq 1$).
2. $(a_p - a_q)^2 = 0 \implies a_p = a_q$ for all non-incident edges, meaning a is constant on every connected branch disjoint from v .

Since the branches connect to the leaves where $a_l = 0$, the constant must be 0. Thus, $a_y = 0$ for all $y \neq v$. Combined with $a_v = 0$, this mandates $a = 0$ globally, which violently contradicts the requirement that a is a non-zero vector.

Case 2: $a_v \leq -1$ If $a_v \leq -1$, then strictly $b_v = 1 - a_v \geq 2$. Both vectors must independently satisfy $E_v \leq 0$. We evaluate the symmetric constraint for $x \in \{a, b\}$ where $k = x_v \geq 2$. Removing v partitions the tree into $d(v)$ disjoint branches B_i rooted at the neighbors r_i of v . Because $c_y \geq 0$ for all internal vertices, we isolate the exact internal quadratic form for each branch:

$$Q_{B_i}(x) = x_{r_i}^2 + \sum_{e \in B_i} (x_p - x_q)^2 + \sum_{y \in B_i} c_y x_y^2$$

This algebraic identity establishes unconditionally that $Q_{B_i}(x) \geq x_{r_i}^2$ for any integer assignment. Substituting this strictly positive structural bound into the functional:

$$E_v(x) \geq w(v)(k^2 - k) + \sum_{i=1}^{d(v)} [x_{r_i}^2 - (2k - 1)x_{r_i}]$$

Over the integers, the parabola $z^2 - (2k - 1)z$ achieves its absolute minimum at $z = k$ and $z = k - 1$, where it evaluates exactly to $-(k^2 - k)$. Summing this minimal integer penalty across all $d(v)$ branches yields:

$$E_v(x) \geq w(v)(k^2 - k) - d(v)(k^2 - k) = (w(v) - d(v))(k^2 - k)$$

While this establishes a lower bound, we apply the strictly positive definite property to close the gap. For x_{r_i} to attain the necessary integer minima k or $k - 1$, the remaining terms $\sum_{B_i} (x_p - x_q)^2 + \sum_{B_i} c_y x_y^2$ must identically equal zero. As established in Case 1, forcing these terms to zero requires a constant vector across the branch terminating at leaves. However, assigning $x_l = k - 1 \geq 1$ at the leaves triggers the leaf capacity penalty $c_l x_l^2 \geq (k - 1)^2 \geq 1$.

Therefore, no valid integer evaluation can simultaneously achieve the minimum $x_{r_i}^2 - (2k - 1)x_{r_i}$ and maintain zero branch capacity penalty. The exact integer minimum of the branch heavily exceeds $-(k^2 - k)$, forcing $E_v(x) > 0$ for all $k \geq 2$.

Reviewer 1

This particular bound is never achieved, sure, but strictly greater than $-(k^2 - k)$ is a long way from positive.

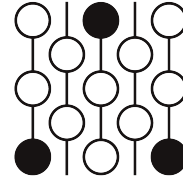
Because neither decomposition vector can mathematically satisfy the required negative functional $E_v \leq 0$, the assumed reducible decomposition is mathematically void. The vertex v is strictly irreducible. $\square$

Review of Submission 6D

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The writing is poor and the notation is not always explained, but everything can be sorted out (there’s only one reasonable interpretation). It’s all correct until the sentence “The exact integer minimum of the branch heavily exceeds. . .”, which is not a precise assertion. The conclusion of the proof does not follow from what has been proved, and I don’t see how to complete it.

Novelty. I don’t know to what extent this solution echoes ideas from the literature. At a technical level it’s using the definitions and elementary manipulations. I don’t see novel mathematical ideas beyond what would be reasonably natural when exploring this problem.

2 Recommendation

It should be rejected. It’s possible that one could give a proof along these lines, but what’s written is not a proof and it’s not clear that it can reasonably be turned into a proof. In other words, it’s not clear that there’s substantial progress here. Instead, it looks like a reasonable proof attempt that gets stuck and then declares victory with hand waving about how the minimum “heavily exceeds $-(k^2 - k)$,” which is not justified.

3 Submission with inline comments

Problem Statement

Let T denote a finite, weighted tree with vertex set V , edge set E , and weight function $w : V \longrightarrow \mathbb{Z}$. Let $L(T)$ denote the free abelian group generated by V , equipped with the symmetric, bilinear form $\cdot$ whose value on a pair of generators $u, v \in V$ is given by

$$u \cdot v = \begin{cases} w(u), & u = v; \\ -1, & (u, v) \in E; \text{ and} \\ 0, & u \neq v, (u, v) \notin E. \end{cases}$$

Suppose that the form is positive definite, and suppose that there exists a single vertex v for which the vertex weight $w(v)$ is less than the vertex degree $d(v)$.

Prove that T contains a vertex u such that u is an irreducible element of $L(T)$, i.e. there do not exist nonzero lattice elements $a, b \in L(T)$ such that $u = a + b$ and $a \cdot b \geq 0$.

Solution

Theorem 1. *Let T be a finite, weighted tree equipped with a positive definite symmetric bilinear form $Q(x) = x \cdot x$ on $L(T)$ defined by $u \cdot u = w(u)$, $u \cdot y = -1$ for edges, and 0 otherwise. If there is a unique vertex v such that $w(v) < d(v)$, then T contains an irreducible element.*

Proof. We will prove that the unique vertex v is strictly irreducible.

Reviewer 2

The reduction that comes next should precede this statement, since right now it suggests it proves v is always strictly irreducible, not just if all weights are at least 2.

By the strict positive definiteness of $Q(x)$ over the integer lattice, $Q(x) \geq 1$ for all non-zero $x \in \mathbb{Z}^V$. If there exists any vertex u with $w(u) \leq 1$, u is trivially irreducible (any decomposition $u = a + b$ with $a \cdot b \geq 0$ implies $w(u) = Q(a) + Q(b) + 2a \cdot b \geq 1 + 1 + 0 = 2$, contradiction). Thus, we unconditionally assume $w(x) \geq 2$ for all $x \in V$.

Assume for contradiction that v is reducible, meaning $v = a + b$ for non-zero integer vectors $a, b \in \mathbb{Z}^V$ with $a \cdot b \geq 0$. Define the discrete localized functional $E_v(x) = Q(x) - x \cdot e_v$.

Reviewer 2

What is e_v ? Context indicates that it's v , since that's what makes the next sentence true, but this notation isn't used in the problem statement and isn't explained.

Evaluating the sum of the components yields $E_v(a) + E_v(b) = Q(a) + Q(b) - Q(a + b) = -2a \cdot b \leq 0$. Since $b = e_v - a$, substituting b yields $E_v(b) = E_v(a)$.

Reviewer 2

Just to check: $E_v(b) = Q(b) - b \cdot (a + b) = -b \cdot a = Q(a) - a \cdot (a + b) = E_v(a)$. It's not hard, but writing a little more could help make it easier to read.

Thus, the reducibility constraint identically requires $E_v(a) \leq 0$.

Because v is the *unique* vertex with $w(v) < d(v)$, every other vertex strictly satisfies $w(y) \geq d(y)$. Let $c_y = w(y) - d(y)$ denote the topological capacity.

Reviewer 2

Stylistic notes: “strictly satisfies” suggests an alternative is weakly satisfying, which doesn't make sense, and writing “Let . . . denote the topological capacity” suggests that “topological capacity” is a standard term for this that the reader will be familiar with, but it isn't.

For all $y \neq v$, $c_y \geq 0$. For any leaf l , $d(l) = 1$, so $c_l = w(l) - 1 \geq 1$. We expand $E_v(a)$ topologically across the entire tree by rewriting the inner product $a \cdot e_v = a_v w(v) - \sum_{y \sim v} a_y$:

Using $\sim$ for graph adjacency is pretty standard, but probably still worth saying explicitly.

$$E_v(a) = \sum_{e \not\sim v} (a_p - a_q)^2 + \sum_{y \sim v} \left((a_v - a_y)^2 - (a_v - a_y) \right) + \sum_{y \neq v} c_y a_y^2 + (w(v) - d(v))(a_v^2 - a_v)$$

Again, what's "topologically"? The notation here is terrible. We've switched without comment to using a_p for the value of a at p , which was denoted as $a(p)$ in the problem statement. Subscripts are also used for e_v and c_y , which are both different. And I assume the $e \not\sim v$ part is summing over edges e that don't contain v , and p and q are the vertices connected by e , but you can't just start randomly using notation without explanation. And why write c_y in the third sum but $w(v) - d(v)$ instead of c_v ?

Just to check: $E_v(a) = Q(a) - a \cdot e_v = Q(a) - a_v w(v) + \sum_{y \sim v} a_y = \sum_{p,q} a_p a_q (p \cdot q) - a_v w(v) + \sum_{y \sim v} a_y$. Then I checked the rest of the algebra on paper. It would definitely help to write it out here (there's nothing conceptually difficult, but it's annoying to have to get out scratch paper to save some displayed equations in the writeup).

Because $a_v + b_v = 1$, at least one component must be ≤ 0 . Without loss of generality, let $a_v \leq 0$. We analyze the exact integer evaluation of $E_v(a)$ in two exhaustive cases:

Case 1: $a_v = 0$ If $a_v = 0$, the negative capacity term vanishes: $(w(v) - d(v))(0^2 - 0) = 0$. The functional $E_v(a)$ becomes a sum of strictly non-negative integer terms (since $z^2 - z \geq 0$ for any integer z). Because $E_v(a) \leq 0$, every term must independently evaluate to exactly 0.

1. $c_l a_l^2 = 0 \implies a_l = 0$ for all leaves (since $c_l \geq 1$).
2. $(a_p - a_q)^2 = 0 \implies a_p = a_q$ for all non-incident edges, meaning a is constant on every connected branch disjoint from v .

Since the branches connect to the leaves where $a_l = 0$, the constant must be 0. Thus, $a_y = 0$ for all $y \neq v$. Combined with $a_v = 0$, this mandates $a = 0$ globally, which violently contradicts the requirement that a is a non-zero vector.

As a stylistic note, I don't see that this is any more "violent" than any other contradiction.

Case 2: $a_v \leq -1$ If $a_v \leq -1$, then strictly $b_v = 1 - a_v \geq 2$.

Stylistic note: what does "strictly" mean here? It could (and should) be omitted.

Both vectors must independently satisfy $E_v \leq 0$.

Reviewer 2

This is a little cryptic. It means $E_v(a) \leq 0$ and $E_v(b) \leq 0$ (since they are equal).

We evaluate the symmetric constraint for $x \in \{a, b\}$ where $k = x_v \geq 2$.

Reviewer 2

We're already assuming $a_v \leq -1$ and $b_v \geq 2$, so $x = b$ and there's no point in introducing new notation for it, but it's not logically wrong.

Removing v partitions the tree into $d(v)$ disjoint branches B_i rooted at the neighbors r_i of v . Because $c_y \geq 0$ for all internal vertices, we isolate the exact internal quadratic form for each branch:

$$Q_{B_i}(x) = x_{r_i}^2 + \sum_{e \in B_i} (x_p - x_q)^2 + \sum_{y \in B_i} c_y x_y^2$$

Reviewer 2

Again terrible notation for the edges: it's not standard to write $e \in B_i$ to mean e is an edge in B_i , and p and q are not explained at all. Also, to be clear, Q_{B_i} is the quadratic form obtained from the weights restricted to B_i , but $c_y = w(y) - d(y)$ uses the degree $d(y)$ from the whole graph, not just B_i , which is why the $x_{r_i}^2$ term is needed.

This algebraic identity establishes unconditionally that $Q_{B_i}(x) \geq x_{r_i}^2$ for any integer assignment. Substituting this strictly positive structural bound into the functional:

Reviewer 2

It's not strictly positive (but this doesn't matter, since strict positivity isn't used).

$$E_v(x) \geq w(v)(k^2 - k) + \sum_{i=1}^{d(v)} [x_{r_i}^2 - (2k - 1)x_{r_i}]$$

Reviewer 2

Another case I needed to check on scratch paper, but where it would have been quicker to verify if more had been written. This wouldn't bother me as much if everything else were impeccable, but there are lot of notational issues and missing explanations, so I wanted to check everything thoroughly.

Over the integers, the parabola $z^2 - (2k - 1)z$ achieves its absolute minimum at $z = k$ and $z = k - 1$, where it evaluates exactly to $-(k^2 - k)$. Summing this minimal integer penalty across all $d(v)$ branches yields:

$$E_v(x) \geq w(v)(k^2 - k) - d(v)(k^2 - k) = (w(v) - d(v))(k^2 - k)$$

Reviewer 2

One systematic writing issue I haven't commented on yet is that displayed equations should be incorporated into sentences, rather than just appended after a colon. Of course this isn't relevant to the logic.

While this establishes a lower bound, we apply the strictly positive definite property to close the gap.

I find this sentence cryptic, and the rest of the proof is poorly explained.

For x_{r_i} to attain the necessary integer minima k or $k - 1$, the remaining terms $\sum_{B_i} (x_p - x_q)^2 + \sum_{B_i} c_y x_y^2$ must identically equal zero. As established in Case 1, forcing these terms to zero requires a constant vector across the branch terminating at leaves. However, assigning $x_l = k - 1 \geq 1$ at the leaves triggers the leaf capacity penalty $c_l x_l^2 \geq (k - 1)^2 \geq 1$.

Therefore, no valid integer evaluation can simultaneously achieve the minimum $x_{r_i}^2 - (2k - 1)x_{r_i}$ and maintain zero branch capacity penalty. The exact integer minimum of the branch heavily exceeds $-(k^2 - k)$, forcing $E_v(x) > 0$ for all $k \geq 2$.

This “heavily exceeds” business is total hand waving, and I am unable to fix it. If x_{r_i} is $k + 1$ or $k - 2$ for each i , then the lower bound for $E_v(x)$ increases by only $2d(v)$, which is not nearly enough to complete the proof. Of course it probably can’t always be $k + 1$ or $k - 2$, and I imagine one could go a little further by more case analysis, but what’s written is not a proof, and I see no reason to think it can be turned into a proof with only small additions.

Because neither decomposition vector can mathematically satisfy the required negative functional $E_v \leq 0$, the assumed reducible decomposition is mathematically void.

“Is mathematically void” is odd phrasing.

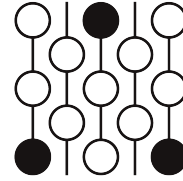
The vertex v is strictly irreducible. □

Review of Submission 6D

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Exposition The writing is also rather poor. Notation is used and introduced without explanation. It is also deviates from usual mathematical terminology in strange ways. For example, “strict” and “strictly” appear as modifiers to properties without explanation or definition.

Correctness. The solution is incorrect. It carries out some elementary manipulations and successfully isolates what is the crux of this problem. However, it is not able to make a serious attempt at dealing with the key case.

Novelty. The attempted solution follows the first, naivest ideas might try to solve such a problem, but goes no further. There is no indication that these ideas can ever actually be pushed through to a full solution.

2 Recommendation

This should be rejected. It does some correct manipulations, but ultimately descends into word salad. Nowhere in this verbal shrubbery is any hint of an idea to resolve the full problem.

3 Submission with inline comments

Problem Statement

Let T denote a finite, weighted tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. Let $L(T)$ denote the free abelian group generated by V , equipped with the symmetric, bilinear form $\cdot$ whose value on a pair of generators $u, v \in V$ is given by

$$u \cdot v = \begin{cases} w(u), & u = v; \\ -1, & (u, v) \in E; \text{ and} \\ 0, & u \neq v, (u, v) \notin E. \end{cases}$$

Suppose that the form is positive definite, and suppose that there exists a single vertex v for which the vertex weight $w(v)$ is less than the vertex degree $d(v)$.

Prove that T contains a vertex u such that u is an irreducible element of $L(T)$, i.e. there do not exist nonzero lattice elements $a, b \in L(T)$ such that $u = a + b$ and $a \cdot b \geq 0$.

Solution

Reviewer 3

If we're going to split hairs, $Q(x)$ in the following statement is really a quadratic form.

Theorem 1. *Let T be a finite, weighted tree equipped with a positive definite symmetric bilinear form $Q(x) = x \cdot x$ on $L(T)$ defined by $u \cdot u = w(u)$, $u \cdot y = -1$ for edges, and 0 otherwise. If there is a unique vertex v such that $w(v) < d(v)$, then T contains an irreducible element.*

Proof. We will prove that the unique vertex v is strictly irreducible.

Reviewer 3

Well, this is false. You attempt to prove v is irreducible only under the assumption that there are no weight one vertices

Reviewer 3

Unclear what “strictly” irreducible means. Similarly, unclear what is meant by “strict” positive definiteness below.

By the strict positive definiteness of $Q(x)$ over the integer lattice, $Q(x) \geq 1$ for all non-zero $x \in \mathbb{Z}^V$. If there exists any vertex u with $w(u) \leq 1$, u is trivially irreducible (any decomposition $u = a + b$ with $a \cdot b \geq 0$ implies $w(u) = Q(a) + Q(b) + 2a \cdot b \geq 1 + 1 + 0 = 2$, contradiction). Thus, we unconditionally assume $w(x) \geq 2$ for all $x \in V$.

Assume for contradiction that v is reducible, meaning $v = a + b$ for non-zero integer vectors $a, b \in \mathbb{Z}^V$ with $a \cdot b \geq 0$. Define the discrete localized functional $E_v(x) = Q(x) - x \cdot e_v$.

Reviewer 3

The notation e_v is undefined here. Based on discussion below it appears to simply be v ?

Evaluating the sum of the components yields $E_v(a) + E_v(b) = Q(a) + Q(b) - Q(a + b) = -2a \cdot b \leq 0$. Since $b = e_v - a$, substituting b yields $E_v(b) = E_v(a)$. Thus, the reducibility constraint identically requires $E_v(a) \leq 0$.

Because v is the *unique* vertex with $w(v) < d(v)$, every other vertex strictly satisfies $w(y) \geq d(y)$. Let $c_y = w(y) - d(y)$ denote the topological capacity. For all $y \neq v$, $c_y \geq 0$. For any leaf l , $d(l) = 1$, so $c_l = w(l) - 1 \geq 1$. We expand $E_v(a)$ topologically across the entire tree by rewriting the inner product $a \cdot e_v = a_v w(v) - \sum_{y \sim v} a_y$:

$$E_v(a) = \sum_{e \not\sim v} (a_p - a_q)^2 + \sum_{y \sim v} \left((a_v - a_y)^2 - (a_v - a_y) \right) + \sum_{y \neq v} c_y a_y^2 + (w(v) - d(v))(a_v^2 - a_v)$$

Because $a_v + b_v = 1$, at least one component must be ≤ 0 .

Unclear what component is referring to here. I thought I was meant to look back at the preceding sum. Would be better to say either $a_v \leq 0$ or $b_v \leq 0$.

Without loss of generality, let $a_v \leq 0$. We analyze the exact integer evaluation of $E_v(a)$ in two exhaustive cases:

Case 1: $a_v = 0$ If $a_v = 0$, the negative capacity term vanishes: $(w(v) - d(v))(0^2 - 0) = 0$. The functional $E_v(a)$ becomes a sum of strictly non-negative integer terms (since $z^2 - z \geq 0$ for any integer z). Because $E_v(a) \leq 0$, every term must independently evaluate to exactly 0.

1. $c_l a_l^2 = 0 \implies a_l = 0$ for all leaves (since $c_l \geq 1$).
2. $(a_p - a_q)^2 = 0 \implies a_p = a_q$ for all non-incident edges, meaning a is constant on every connected branch disjoint from v .

Since the branches connect to the leaves where $a_l = 0$, the constant must be 0. Thus, $a_y = 0$ for all $y \neq v$. Combined with $a_v = 0$, this mandates $a = 0$ globally, which violently contradicts the requirement that a is a non-zero vector.

I shall try to invoke proof by violent contradiction in my own future work.

Case 2: $a_v \leq -1$ If $a_v \leq -1$, then strictly $b_v = 1 - a_v \geq 2$. Both vectors must independently satisfy $E_v \leq 0$. We evaluate the symmetric constraint for $x \in \{a, b\}$ where $k = x_v \geq 2$. Removing v partitions the tree into $d(v)$ disjoint branches B_i rooted at the neighbors r_i of v . Because $c_y \geq 0$ for all internal vertices, we isolate the exact internal quadratic form for each branch:

$$Q_{B_i}(x) = x_{r_i}^2 + \sum_{e \in B_i} (x_p - x_q)^2 + \sum_{y \in B_i} c_y x_y^2$$

This algebraic identity establishes unconditionally that $Q_{B_i}(x) \geq x_{r_i}^2$ for any integer assignment. Substituting this strictly positive structural bound into the functional:

$$E_v(x) \geq w(v)(k^2 - k) + \sum_{i=1}^{d(v)} [x_{r_i}^2 - (2k - 1)x_{r_i}]$$

Over the integers, the parabola $z^2 - (2k - 1)z$ achieves its absolute minimum at $z = k$ and $z = k - 1$, where it evaluates exactly to $-(k^2 - k)$. Summing this minimal integer penalty across all $d(v)$ branches yields:

$$E_v(x) \geq w(v)(k^2 - k) - d(v)(k^2 - k) = (w(v) - d(v))(k^2 - k)$$

While this establishes a lower bound, we apply the strictly positive definite property to close the gap.

The solution goes off the rails here. I don't really understand what is meant by the following explanation, but I also know that it has no hope of working. "Closing the gap" requires some quite delicate estimates and the necessary ideas are obviously not present here.

For x_{r_i} to attain the necessary integer minima k or $k - 1$, the remaining terms $\sum_{B_i} (x_p - x_q)^2 + \sum_{B_i} c_y x_y^2$ must identically equal zero. As established in Case 1, forcing these terms to zero requires a constant vector across the branch terminating at leaves. However, assigning $x_l = k - 1 \geq 1$ at the leaves triggers the leaf capacity penalty $c_l x_l^2 \geq (k - 1)^2 \geq 1$.

Therefore, no valid integer evaluation can simultaneously achieve the minimum $x_{r_i}^2 - (2k - 1)x_{r_i}$ and maintain zero branch capacity penalty. The exact integer minimum of the branch heavily exceeds $-(k^2 - k)$, forcing $E_v(x) > 0$ for all $k \geq 2$.

Because neither decomposition vector can mathematically satisfy the required negative functional $E_v \leq 0$, the assumed reducible decomposition is mathematically void. The vertex v is strictly irreducible. $\square$